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1  # Enterprise Cloud Infrastructure RFP - Requirements Specification
2  # Issued by: Fairbanks North Star Borough IT Department
3  # Document Version: 3.2 | Date: June 2026
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5  ## Section 1: General Requirements
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7  ### 1.1 Vendor Qualifications
8  REQ-001: The vendor must demonstrate a minimum of 5 years of experience providing cloud
9  infrastructure services to government or public sector organizations.
10 REQ-002: The vendor must hold current SOC 2 Type II certification with audit report
11 dated within the last 12 months.
12 REQ-003: The vendor must maintain FedRAMP Moderate authorization or equivalent security
13 certification for all proposed cloud services.
14 REQ-004: The vendor must provide at least three (3) references from comparable public
15 sector clients with contact information for project managers.
16 REQ-005: The vendor must demonstrate financial stability through audited financial
17 statements for the last three fiscal years.
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19 ### 1.2 Project Management
20 REQ-006: The vendor must assign a dedicated project manager who holds PMP or equivalent
21 certification and is available during Alaska business hours (AKST).
22 REQ-007: The vendor must provide a detailed project plan within 15 business days of
23 contract execution, including milestones, deliverables, and resource allocation.
24 REQ-008: The vendor must conduct weekly status meetings with Borough IT leadership and
25 provide written status reports within 24 hours of each meeting.
26 REQ-009: The vendor must maintain a risk register and present risk mitigation updates at
27 each monthly steering committee meeting.
28 REQ-010: The vendor must provide a dedicated escalation path with guaranteed response
29 times: Critical (1 hour), High (4 hours), Medium (8 hours), Low (24 hours).
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31 ## Section 2: Technical Architecture Requirements
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33 ### 2.1 Cloud Platform
34 REQ-011: All proposed services must be hosted within the continental United States or US
35 sovereign cloud regions.
36 REQ-012: The solution must support hybrid cloud architecture allowing integration with
37 the Borough's existing on-premises VMware environment.
38 REQ-013: The platform must provide 99.99% uptime SLA for production workloads, measured
39 monthly.
40 REQ-014: The solution must support automatic failover to a geographically separate
41 region within 15 minutes of primary region failure.
42 REQ-015: The platform must support both Windows Server 2019/2022 and Linux (RHEL 8+,
43 Ubuntu 22.04+) workloads.
44 REQ-016: The solution must provide container orchestration capabilities (Kubernetes)
45 with managed control plane.
46 REQ-017: The platform must support Infrastructure as Code (IaC) using Terraform, Bicep,
47 or ARM templates.
48 REQ-018: The solution must provide serverless computing options for event-driven
49 workloads.
50 REQ-019: The platform must support GPU-accelerated instances for GIS processing and data
51 analytics workloads.
52 REQ-020: The solution must provide managed database services supporting SQL Server,
53 PostgreSQL, and MongoDB.
54
55 ### 2.2 Networking
56 REQ-021: The solution must provide private connectivity from Borough facilities to the
57 cloud platform via dedicated circuit (minimum 1 Gbps).
58 REQ-022: The platform must support virtual network segmentation with micro-segmentation
59 capabilities.
60 REQ-023: The solution must include managed DDoS protection for all public-facing
61 endpoints.
62 REQ-024: The platform must support IPv4 and IPv6 dual-stack networking.
63 REQ-025: The solution must provide DNS management with DNSSEC support.
64 REQ-026: The platform must include a web application firewall (WAF) with OWASP Top 10
65 rule sets.
66 REQ-027: The solution must support site-to-site VPN as a backup to dedicated
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connectivity with automatic failover.

43 REQ-028: The platform must provide network traffic analytics and flow logging for all virtual networks.

44 REQ-029: The solution must support traffic inspection and intrusion detection/prevention (IDS/IPS) for east-west traffic.

45 REQ-030: The platform must provide content delivery network (CDN) capabilities for public-facing applications.

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47 ### 2.3 Storage

48 REQ-031: The solution must provide object storage with 99.999999999% (11 nines) durability.

49 REQ-032: The platform must support tiered storage (hot, cool, archive) with automated lifecycle management policies.

50 REQ-033: The solution must provide block storage with guaranteed IOPS and throughput SLAs.

51 REQ-034: The platform must support file storage (SMB 3.0 and NFS 4.1) for legacy application compatibility.

52 REQ-035: The solution must provide immutable storage capabilities for records retention compliance.

53 REQ-036: All data at rest must be encrypted using AES-256 or equivalent, with customer-managed key options.

54 REQ-037: The platform must support cross-region replication for disaster recovery with configurable RPO.

55 REQ-038: The solution must provide data deduplication and compression to optimize storage costs.

56 REQ-039: The platform must support large file transfers (up to 100 TB) via physical appliance or high-speed transfer tools.

57 REQ-040: The solution must provide backup and restore capabilities with granular recovery options (file-level, volume-level, application-consistent).

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59 ## Section 3: Security Requirements

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61 ### 3.1 Identity and Access Management

62 REQ-041: The solution must integrate with the Borough's existing Microsoft Entra ID (Azure AD) for single sign-on (SSO).

63 REQ-042: The platform must support multi-factor authentication (MFA) for all administrative access.

64 REQ-043: The solution must provide role-based access control (RBAC) with principle of least privilege enforcement.

65 REQ-044: The platform must support Privileged Access Management (PAM) with just-in-time (JIT) access capabilities.

66 REQ-045: The solution must provide comprehensive identity governance including access reviews and certification campaigns.

67 REQ-046: The platform must support conditional access policies based on user location, device compliance, and risk level.

68 REQ-047: The solution must provide service account management with automatic credential rotation.

69 REQ-048: The platform must support federated identity for inter-agency collaboration without credential sharing.

70 REQ-049: The solution must log all authentication and authorization events with tamper-evident audit trails.

71 REQ-050: The platform must support hardware security keys (FIDO2) as an MFA option for high-privilege accounts.

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73 ### 3.2 Data Protection

74 REQ-051: All data in transit must be encrypted using TLS 1.3 or higher.

75 REQ-052: The solution must provide data loss prevention (DLP) capabilities for sensitive data classification and protection.

76 REQ-053: The platform must support customer-managed encryption keys (CMEK) stored in a FIPS 140-2 Level 3 validated HSM.

77 REQ-054: The solution must provide data classification and labeling capabilities integrated with Microsoft Information Protection.

78 REQ-055: The platform must support secure key management with automatic rotation and revocation capabilities.

79 REQ-056: The solution must provide database encryption (TDE) for all managed database services.

80 REQ-057: The platform must support tokenization for sensitive fields in application
databases.

81 REQ-058: The solution must provide certificate management with automated renewal and
deployment.

82 REQ-059: The platform must support secure secrets management (API keys, connection
strings, passwords) with audit logging.

83 REQ-060: The solution must provide data masking capabilities for non-production
environments.

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85 ### 3.3 Threat Protection

86 REQ-061: The solution must provide 24/7 Security Operations Center (SOC) monitoring with
threat detection and response.

87 REQ-062: The platform must include cloud-native SIEM capabilities with at least 90 days
of hot storage for security logs.

88 REQ-063: The solution must provide automated threat response (SOAR) for common attack
patterns.

89 REQ-064: The platform must include vulnerability scanning for all deployed workloads
with weekly automated scans.

90 REQ-065: The solution must provide endpoint detection and response (EDR) for all virtual
machines.

91 REQ-066: The platform must include email security (anti-phishing, anti-malware, safe
attachments) for Borough email.

92 REQ-067: The solution must provide container security scanning for vulnerabilities in
images before deployment.

93 REQ-068: The platform must include API security monitoring and anomaly detection.

94 REQ-069: The solution must provide deception technology (honeypots/honeytokens) as an
early warning system.

95 REQ-070: The platform must support threat intelligence feed integration for proactive
threat hunting.

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97 ## Section 4: Compliance and Governance

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99 ### 4.1 Regulatory Compliance

100 REQ-071: The solution must comply with Alaska state data privacy laws (AS 45.48).

101 REQ-072: The platform must support CJIS Security Policy compliance for law enforcement
data.

102 REQ-073: The solution must comply with HIPAA/HITECH for health-related data processed by
Borough health services.

103 REQ-074: The platform must support PCI DSS compliance for payment card processing
environments.

104 REQ-075: The solution must comply with ADA Section 508 for all user-facing management
interfaces.

105 REQ-076: The platform must provide compliance reporting dashboards with evidence
collection for audit preparation.

106 REQ-077: The solution must support data sovereignty requirements ensuring all Borough
data remains within US jurisdiction.

107 REQ-078: The platform must provide automated compliance posture assessment against CIS
benchmarks.

108 REQ-079: The solution must support records retention policies aligned with Borough
records management schedule.

109 REQ-080: The platform must provide chain of custody documentation for all data handling
processes.

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111 ### 4.2 Governance

112 REQ-081: The solution must provide resource tagging enforcement with mandatory tags for
cost center, environment, owner, and data classification.

113 REQ-082: The platform must support policy-as-code for automated governance enforcement.

114 REQ-083: The solution must provide cost management and optimization recommendations with
monthly reporting.

115 REQ-084: The platform must support resource quotas and spending limits with automated
alerts at 80% and 95% thresholds.

116 REQ-085: The solution must provide change management workflows with approval gates for
production changes.

117 REQ-086: The platform must support multi-environment management (dev, test, staging,
production) with promotion workflows.

118 REQ-087: The solution must provide asset inventory with automatic discovery and

classification.

119 REQ-088: The platform must support sustainability reporting including carbon footprint estimates for cloud usage.

120 REQ-089: The solution must provide license management tracking for all software deployed on the platform.

121 REQ-090: The platform must support organizational hierarchy mapping for cost allocation and access control inheritance.

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123 ## Section 5: Operations and Support

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125 ### 5.1 Monitoring and Observability

126 REQ-091: The solution must provide comprehensive monitoring covering infrastructure, application, and business metrics.

127 REQ-092: The platform must support custom dashboards with real-time visualization and configurable refresh intervals.

128 REQ-093: The solution must provide alerting with multiple notification channels (email, SMS, Teams, PagerDuty).

129 REQ-094: The platform must support distributed tracing for microservices architectures.

130 REQ-095: The solution must provide log aggregation with structured search and minimum 1-year retention.

131 REQ-096: The platform must support synthetic monitoring for external service availability verification.

132 REQ-097: The solution must provide capacity planning tools with predictive analytics for resource utilization.

133 REQ-098: The platform must support custom metric collection from Borough-developed applications.

134 REQ-099: The solution must provide Service Level Objective (SLO) tracking with error budget monitoring.

135 REQ-100: The platform must support AIOps capabilities for anomaly detection and root cause analysis.

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137 ### 5.2 Disaster Recovery and Business Continuity

138 REQ-101: The solution must support Recovery Time Objective (RTO) of 4 hours for Tier 1 (critical) applications.

139 REQ-102: The platform must support Recovery Point Objective (RPO) of 1 hour for Tier 1 applications.

140 REQ-103: The solution must provide automated DR failover testing without impacting production workloads.

141 REQ-104: The platform must support runbook automation for disaster recovery procedures.

142 REQ-105: The solution must provide regular DR drill reports with findings and remediation tracking.

143 REQ-106: The platform must support backup immutability to protect against ransomware attacks.

144 REQ-107: The solution must provide cross-region DR with configurable replication for each application tier.

145 REQ-108: The platform must support application-consistent snapshots for complex multi-tier applications.

146 REQ-109: The solution must provide DR orchestration with dependency-aware recovery sequencing.

147 REQ-110: The platform must support bare-metal recovery for critical infrastructure components.

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149 ### 5.3 Support and SLA

150 REQ-111: The vendor must provide 24/7/365 technical support for critical issues.

151 REQ-112: The vendor must assign a Technical Account Manager (TAM) familiar with Borough infrastructure.

152 REQ-113: The vendor must provide a self-service knowledge base and documentation portal.

153 REQ-114: The vendor must offer training for Borough IT staff (minimum 40 hours initial training, 16 hours annual refresher).

154 REQ-115: The vendor must provide quarterly business reviews with capacity planning and optimization recommendations.

155 REQ-116: The vendor must maintain a service roadmap with at least 12-month visibility and advance notice of breaking changes.

156 REQ-117: The vendor must provide incident post-mortem reports within 5 business days of any P1/P2 incident.

157 REQ-118: The vendor must support the Borough's ITSM tool (ServiceNow) for ticket

integration.

158 REQ-119: The vendor must provide a dedicated support channel (not shared queue) with
named support engineers.

159 REQ-120: The vendor must offer professional services for architecture reviews,
migrations, and optimization projects at pre-negotiated rates.

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161 ## Section 6: Migration Requirements

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163 ### 6.1 Migration Planning

164 REQ-121: The vendor must conduct a comprehensive discovery and assessment of all Borough
workloads within 30 days of contract start.

165 REQ-122: The vendor must provide a migration plan categorizing workloads into rehost,
replatform, refactor, or retire.

166 REQ-123: The vendor must demonstrate zero-downtime migration capability for critical
applications.

167 REQ-124: The vendor must provide rollback procedures for each migration wave with tested
recovery steps.

168 REQ-125: The vendor must support parallel-run periods of at least 30 days for critical
applications before cutover.

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170 ### 6.2 Data Migration

171 REQ-126: The vendor must provide secure data transfer mechanisms with integrity
verification (checksums) for all migrated data.

172 REQ-127: The vendor must support database migration with schema conversion assistance
for incompatible databases.

173 REQ-128: The vendor must demonstrate data validation procedures ensuring 100% data
integrity post-migration.

174 REQ-129: The vendor must provide progress tracking and reporting during migration with
daily status updates.

175 REQ-130: The vendor must support offline data transfer via encrypted physical appliances
for datasets exceeding 10 TB.

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177 ## Section 7: Financial Requirements

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179 ### 7.1 Pricing

180 REQ-131: The vendor must provide transparent pricing with no hidden fees or surcharges.

181 REQ-132: The vendor must offer committed-use discounts for 1-year and 3-year terms.

182 REQ-133: The vendor must provide a detailed cost model for the Borough's projected
workload (provided in Appendix B).

183 REQ-134: The vendor must cap annual price increases at no more than 5% for the duration
of the contract.

184 REQ-135: The vendor must offer a cost-neutral pilot period of 90 days to validate the
platform before full commitment.

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